# COAL - Assignment 02

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## Q1

```
mov ax , 3
mov bx , ax

shl ax , 4
mov cx , bx

shl cx , 3
shl bx , 1

add ax , cx
add ax , bx
```

## Q2

```
; Right
  mov ecx , 4
R1:
  shr BYTE PTR [dval + 3], 1
  rcr BYTE PTR [dval + 2], 1
```

```
rcr BYTE PTR [dval + 1], 1
rcr BYTE PTR [dval], 1
loop R1

; Left
mov ecx, 4
L1:
shl BYTE PTR [dval], 1
rcl BYTE PTR [dval+1], 1
rcl BYTE PTR [dval +2], 1
rcl BYTE PTR [dval +3], 1
loop L1
```

## Q3

```
; Right
mov ecx, 2
R1:
shr WORD PTR [dval +2], 1
rcr WORD PTR [dval], 1
loop

; Left
mov ecx, 2
L1:
    shl WORD PTR [dval], 1
    rcl WORD PTR [dval +2], 1
    loop L1
```

# Q4

```
INCLUDE Irvine32.inc

.data
    BSeconds BYTE ?
    BMinutes BYTE ?

.code
main PROC
    mov dx, 0101100111010110b

    ; (a) Seconds
    mov al, dl
    and al, 00011111b
    mov Bseconds, al

    ; (b) Minutes
    mov ax, dx
    shr ax, 5
    and al, 00111111b
    mov BMinutes, al

    exit
main ENDP
end main
```

## Q5

- Step 1: AX = 7F80 + 90EA = 006A, CF = 1
- Step 2: DX = 9A3F + 1 + 1 = 9A41, CF = 0
- Step 3: AX = 006A + 8120 = 818A, CF = 0
- Step 4: DX = 9A41 + FFFF + 0 = 9A40, CF = 1

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- Final Ans: DX = 9A40h, AX = 818Ah, CF = 1

## Q6

```
test  ax , ax      ; (a)
test  bx , 1       ; (b)
test  dx , 8000h   ; (c)
test  dx , 8000h   ; (d)
test  bl , 0FFh    ; (e)
```

## Q7

DX = 5 + 10h + 1 => **DX = 0016h (22)**

# Q8

## LOGIC ERRORS:

1) esi / edi must hold addresses, not values of val1, val2
2) The bytes are processed from least significant upward so dec must be inc.
3) The result must be stored in result which needs its own pointer
4) loop uses ecx, not cx.

## CORRECTED CODE:

```
.code
main PROC
        mov ecx, 8
        mov esi, OFFSET val1
        mov edi, OFFSET val2
        mov ebx, OFFSET result
        clc
top:
        mov al, [esi]
        sbb al, [edi]
        mov [ebx], al
        inc esi
        inc edi
        inc ebx
        loop top
```

```
    exit
main ENDP
end main
```

## Q9

```
INCLUDE Irvine32.inc

.data
    prompt BYTE "Enter marks (0-100): ", 0
    msg BYTE "Grade: ", 0

.code
main PROC
    mov edx, OFFSET prompt
    call WriteString
    call ReadInt

    cmp eax, 80
    jge gA
    cmp eax, 60
    jge gB
    cmp eax, 40
    jge gC
    mov bl, 'F'
    jmp show
```

```
gA:
        mov  bl, 'A'
        jmp  show
gB:
        mov  bl, 'B'
        jmp  show
gC:
        mov  bl, 'C'
        jmp  show

show:
        mov  edx , OFFSET msg
        call WriteString
        mov  al, bl
        call WriteChar
        exit
main    ENDP
end  main
```

Q10

- push 3, push 7, then pop eax gives EAX = 7. The stack holds [3]
- push 10, then pop ebx gives EBX = 10. pop eax gives EAX = 3

- Final: EAX = 3, EBX = 10, stack empty

## Q11

```
.code
main PROC
    mov  eax, num1
    cmp  eax, num2
    jle  elseif

    mov  ebx, num3
    cmp  ebx, num4
    je   elseif

    add  eax, ebx
    mov  result, eax
    jmp  done

elseif:
    mov  eax, num2
    cmp  eax, num4
    jne  else1
    sub  eax, num1
    mov  result, eax
    jmp  done
else1:
    mov  result, 0
done:
    exit
main ENDP
end main
```

Q12

```
.code
main PROC

top:
        mov  eax, op1
        cmp  eax, op2
        jg   endw

        cmp  eax, x
        jl   else part
        cmp  eax, y
        jg   elsepart

        add  z, 10
        jmp  next

elsepart:
        sub  z, 10
next:
        dec  op1
        jmp  top
endw:
        exit
main ENDP
end main
```